1. Clinical Question & Core Hypothesis

Clinical Question

From the Ontologist, the core question is why genomic instability (MSH6 dysfunction and high TMB) and measurable immune infiltration do not uniformly translate into effective antitumor immunity or consistent benefit from immunotherapy in bladder cancer. 2) Tier 1 paths used: I built on the MSH6-as-prognostic-factor path because MSH6 links MMR-deficiency to neoantigen generation, and on the TMB path because TMB quantifies mutational/neoantigen burden that typically drives immune recruitment; these two provide the strongest mechanistic backbone for explaining discordance between immune presence and function. 3) Tier 2 support: GARS1 (tumor-intrinsic modifier of immune context), systemic/metabolic state (HDL-C), urine-DNA methylation (epigenetic tumor signatures), MSI (MMR phenotype), and the concept of Immunoscore provide plausible mechanisms that can modulate antigen presentation, neoantigen clonality, and immune exhaustion—thus explaining why TIL presence may be non-protective in some genomic contexts. 4) Why therapy_response as endpoint: The Ontologist highlighted variable immunotherapy outcomes despite high mutational/immune signals; testing a therapy-response proxy (immune checkpoint responsiveness) is the most clinically actionable question emerging from these paths. 5) What I excluded and why: I deliberately excluded cross-cancer or weakly linked items (PSA, stool-based colorectal methylation, SARIFA) from the core hypothesis to keep it mechanistic and focused on bladder-specific genomic–immune interactions; Immunoscore and other Tier2 items were moved to secondary_modifiers to preserve the Focusing Rule limit of max two anchors and one imaging feature?

Working Hypothesis

In bladder cancer, tumors with MSH6 deficiency exhibit MMR dysfunction and high tumor mutational burden (TMB), which drives robust infiltration by tumor-infiltrating lymphocytes (TILs). However, the neoantigen landscape in these tumors is highly subclonal, leading to incomplete and non-uniform antigen presentation across tumor cells. This subclonal architecture undermines effective T cell recognition and sustains T cell dysfunction/exhaustion, such that TIL-high, MSH6-deficient, high-TMB tumors show poorer responses to immune checkpoint blockade (ICB) than TIL-high tumors with intact MSH6 and relatively clonal neoantigen profiles. The two core anchors are: 1) imaging_feature: tumor-infiltrating lymphocyte density, and 2) anchor_genes: MSH6 status and TMB level. Testable predictions include: a) among TIL-high bladder tumors, those with MSH6 deficiency and high TMB will have lower objective response rates and shorter progression-free survival on PD-1/PD-L1 inhibitors than TIL-high tumors with MSH6 proficiency and/or lower TMB; b) this difference will be mediated by higher subclonal neoantigen burden, reduced antigen presentation pathway activity (e.g., HLA/B2M expression), and greater T cell exhaustion marker expression; and c) secondary modifiers (GARS1, HDL-C, urine DNA methylation signatures, MSI status, Immunoscore) will modulate, but not reverse, the direction and magnitude of the effect. This framework remains focused on bladder-specific genomic–immune interactions and uses imaging-informed endpoints (therapy_response) to ensure clinical relevance.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MSH6, TMB
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6-as-prognostic-factor path because MSH6 links MMR-deficiency to neoantigen generation, and on the TMB path because TMB quantifies mutational/neoantigen burden that typically drives immune recruitment; these two provide the strongest mechanistic backbone for explaining discordance between immune presence and function.

TMB: TMB path because TMB quantifies mutational/neoantigen burden that typically drives immune recruitment; these two provide the strongest mechanistic backbone for explaining discordance between immune presence and function.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	GARS1	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	Urine-DNA methylation biomarkers	Supporting modifier from knowledge graph
Secondary	MSI	Supporting modifier from knowledge graph
Secondary	Immunoscore	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

Prediction 3: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.
- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- GARS1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- HDL-C: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.